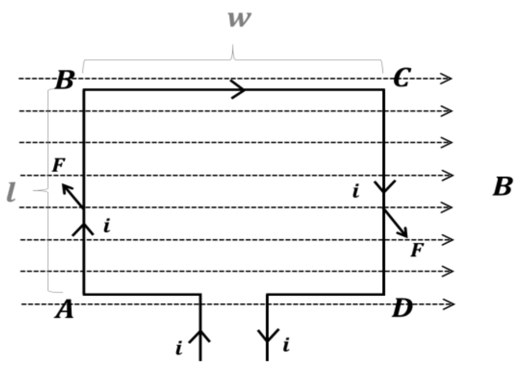
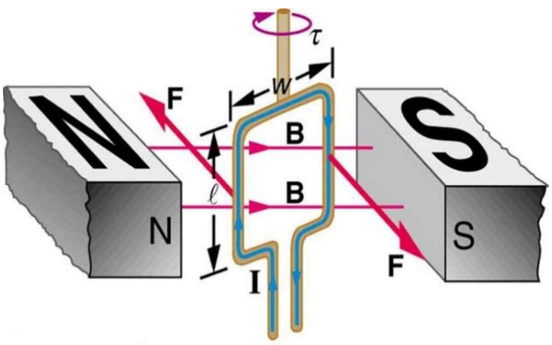
Torque on a Current Carrying Coil in a Magnetic field:

We know that it a current carrying wire is placed in a magnetic field, a magnetic force acts on it. From this phenomenon, when a current carrying loop or coil is placed in a magnetic field, magnetic force and torque produced due to the current flow in the coil.

Let $ABCD$ be very small rectangular loop or coil carrying current placed in a uniform magnetic field B. The plane of the coil is in parallel position to the magnetic field. The loop is attached to a shaft or rod in such a way as it can rotate freely as shown in figure.





As the loop is rectangular –

$$AB = CD = l \qquad \text{(length of the loop)}$$

and $$BC = DA = w \qquad \text{(width of the loop)}$$

Let the current flow, i is clockwise. Now the force applied on the arms –

$$AB \rightarrow \qquad F = ilBsin90 = ilB$$

$$BC \rightarrow \qquad F = iwBsin0 = 0$$

$$CD \rightarrow \qquad F = ilBsin90 = ilB$$

$$DA \rightarrow \qquad F = iwBsin180 = 0$$

According to the right hand rule, the forces acting on AB and CD are equal and opposite to each other and their line of action not being in same straight line, the loop will experience a torque or torsional force; consequently the loop will turn or rotate with respect to the rod. Now – Net torque = Force × perpendicular distance between the forces acting

$$\tau = F \times w$$

$$\tau = ilB \times w = ilwB$$

$$\tau = iAB$$

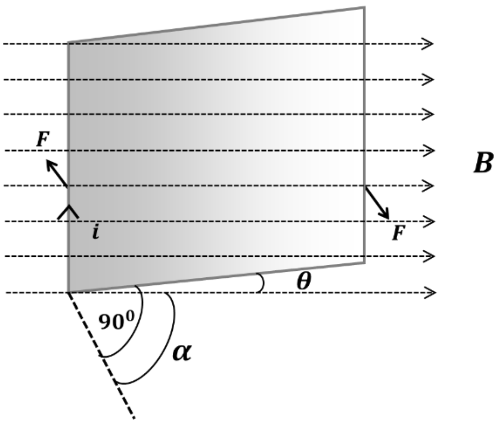
Here, $A = lw$, the area of the loop.

If the coil is made by warpping N number of turns of a wire, then –

$$\tau = NiAB$$

If the plane of the loop or coil makes an angle θ with the magnetic field, then –

$$\boldsymbol{\tau = NiABcos\theta}$$



Again, If the normal of the plane (Area vector, $\vec{A}$) of the loop or coil makes an angle α with the magnetic field, then –

$$\boldsymbol{\tau = NiABsin\alpha}$$

In the vector form it can be written as –

$$\tau = Ni(\vec{A} \times \vec{B})$$

$Ni\vec{A}$ is called the magnetic moment, $\vec{M}$ of the coil. The direction of $\vec{M}$ will be along the direction of $\vec{A}$. Now –

$$\boldsymbol{\tau = \vec{M} \times \vec{B}}$$

Magnetic Moment, $\vec{M}$: *The product of the current of a current carrying coil and area of the coil is called the magnetic moment of that coil.*

Problem-1: *An electron is moving with a speed of* $5 \times 10^7 m/s$ *at right angle to a magnetic field of* $5000G$*. What is the magnetic force on the electron?*

Problem-2: *A coil of* 20 *turns has an area of* $800mm^2$ *and bears a current of* $0.5A$*. It is placed with its plane parallel to magnetic field of intensity* $0.3T$*. Determine the torque and magnetic moment on the coil.*

Problem-3: *Current flowing in a circular coil of radius of* $5 \times 10^{-2}m$ *having number of turns* 100 *is* $1\ ampere$*. If it is placed in a magnetic field of* $1.5 \times 10^{-2} wbm^{-2}$ *at an angle of* 30^0*between normal of the plane of the coil and the magnetic field, what magnitude of torque will be applied on the coil?*

Problem-4: *See the book "Physics for Engineers (Part-2) - Gias Uddin Ahmed"; Page-214*